

# CSCI 112 Assignment – C Language Fundamentals

Points: 10

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## Part A — Lab Instructions

**Submission:** Canvas Quiz — Lab 2

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### Objective

Practice the following topics by writing C code from plain-English instructions:

- `printf` formatting
  - Integer arithmetic and modulo
  - Debugger — breakpoints and Locals panel
  - Casting — implicit and explicit
  - Scope and variable visibility
  - Logical and relational operators
  - Unsigned types and wraparound
  - Prefix and postfix operators
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### Setup

1. Create a new C project in your IDE
  2. Replace the contents of `main.c` with the provided `skeleton.c`
  3. Compile — **zero errors, zero warnings** before you start
  4. Open **Canvas Quiz — Lab 3** in a separate window
  5. Work through sections **1 through 7** in order
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### How this lab works

**Section 1 is the only section with `printf` already written.** You will not write any `printf` calls in sections 2 through 7. All variable values in those sections are observed exclusively through the **debugger Locals panel**.

For every section from 2 to 7: read the plain-English steps, translate each one into a valid line of C code, then use the debugger to observe the result and answer the quiz questions.

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## Debugger — Quick Reference

| Action               | How                                                    |
|----------------------|--------------------------------------------------------|
| Set a breakpoint     | Click to the left of a line number — a red dot appears |
| Start debugging      | F5                                                     |
| Execute one line     | F10 — Step Over                                        |
| Read variable values | <b>Locals panel</b> at the bottom of the screen        |

A variable appears in the Locals panel only **after** the line that declares it has been executed. Press F10 to land **on** a line, then press F10 once more to execute it — then read the Locals panel.

## Section Overview

| Section                                | Your task                                                                                                                        |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| 1 — <b>printf</b>                      | Run the program. Observe the output. Modify one format specifier and observe the change. Answer Q1–Q2.                           |
| 2 — <b>Arithmetic</b>                  | Translate 8 steps into C. Use the debugger to observe results. Answer Q3–Q5.                                                     |
| 3 — <b>Debugger / sizeof</b>           | Translate 10 steps into C. Read every variable from the Locals panel. Answer Q6–Q7.                                              |
| 4 — <b>Casting</b>                     | Translate 11 steps into C. Observe implicit and explicit conversions in Locals. Answer Q8–Q11.                                   |
| 5 — <b>Scope</b>                       | Translate 7 steps into C. Watch <code>val</code> appear, change, and disappear in Locals. Answer Q12–Q13.                        |
| 6 — <b>Logical operators</b>           | Translate 9 steps into C. Record all results from Locals. Answer Q14–Q15.                                                        |
| 7 — <b>Unsigned / prefix / postfix</b> | Translate 11 steps into C. Observe wraparound and the difference between <code>i++</code> and <code>++i</code> . Answer Q16–Q20. |

## PART B — `skeleton.c`

```
#include <stdio.h>
```

```

int main()
{
    /* =====
    SECTION 1 | printf and format specifiers

    This section is already written for you.
    Run the program and observe the full output.
    Then modify the line that uses %.2f:
    change it to %.0f, run again, and observe what changes.
    Answer Q1-Q3 from what you see.
    ===== */

    char name[] = "Grace Hopper";
    int year = 1906;
    float score = 97.5f;
    int rank = 1;

    printf("=====\n");
    printf(" Name: %s\n", name);
    printf(" Year: %d\n", year);
    printf(" Score: %.2f\n", score); /* <-- modify this line for Q1 */
    printf(" Rank: %05d\n", rank);
    printf("=====\n");
    printf("Col1\tCol2\tCol3\n");
    printf("%d\t%d\t%d\n", 1, 2, 3);
    printf("score with %f : %f\n", score);
    printf("score with %.1f : %.1f\n", score);
    printf("score with %.4f : %.4f\n", score);

    /* =====
    SECTION 2 | Integer arithmetic

    Translate each step into one line of C code.
    Place a breakpoint on Step 1.
    Press F10 to step line by line.
    Read values from the Locals panel after each step.
    Answer Q4-Q6 from what you observe.

    Step 1. Declare an integer variable named a with value 7.
    Step 2. Declare an integer variable named b with value 2.
    Step 3. Declare an integer variable named r1 that stores
            the result of a divided by b.
    Step 4. Declare an integer variable named r2 that stores
            the remainder when a is divided by b.
    Step 5. Declare a float variable named fa with value 7.0f.
    ===== */

```

```

Step 6. Declare a float variable named r3 that stores
        the result of fa divided by b.
Step 7. Declare an integer variable named r4 that stores
        the result of 15 divided by 4.
Step 8. Declare an integer variable named r5 that stores
        the remainder when 15 is divided by 4.
===== */

/* --- write your Section 2 code here --- */

/* =====
SECTION 3 | Debugger and sizeof

Translate each step into one line of C code.
Place a breakpoint on Step 1.
Press F10 to step through all lines.
Read ALL values from the Locals panel.
Answer Q7-Q9 from what you observe.

Step 1. Declare an integer variable named p with value 10.
Step 2. Declare an integer variable named q with value 4.
Step 3. Declare an integer variable named d1 that stores
        the sum of p and q.
Step 4. Declare an integer variable named d2 that stores
        the integer division of p by q.
Step 5. Declare an integer variable named d3 that stores
        the remainder of p divided by q.
Step 6. Declare an integer variable named sz_char that
        stores the number of bytes occupied by type char.
        Use the sizeof operator.
Step 7. Declare an integer variable named sz_short that
        stores the number of bytes occupied by type short.
Step 8. Declare an integer variable named sz_int that
        stores the number of bytes occupied by type int.
Step 9. Declare an integer variable named sz_float that
        stores the number of bytes occupied by type float.
Step 10. Declare an integer variable named sz_double that
        stores the number of bytes occupied by type double.
===== */

/* --- write your Section 3 code here --- */

/* =====
SECTION 4 | Casting

```

Translate each step into one line of C code.  
 Place a breakpoint on Step 1.  
 Press F10 to step through all lines.  
 Read values from the Locals panel after each step.  
 Answer Q10-Q13 from what you observe in the Locals panel.

Step 1. Declare an integer variable named `ci` with value 5.  
 Step 2. Declare a float variable named `cf` with value 2.7f.  
 Step 3. Declare an integer variable named `c1` that stores  
           the result of `ci + cf`.  
 Step 4. Declare a float variable named `c2` that stores  
           the result of `ci + cf`.  
 Step 5. Declare an integer variable named `c3` that stores  
           `cf` cast explicitly to `int`.  
           Write the cast operator as: `(int)cf`  
 Step 6. Declare a float variable named `c4` using  
           the expression: `(float)ci / 2`  
 Step 7. Declare a float variable named `c5` using  
           the expression: `(float)(ci / 2)`

Step 8. Declare a char variable named `ch` with value 'A'.  
 Step 9. Declare an integer variable named `ch_int` that  
           stores `ch`.  
 Step 10. Declare a char variable named `ch_lower` that  
           stores `ch + 32`.  
 Step 11. Declare an integer variable named `ch_lower_int`  
           that stores `ch_lower`.

===== \*/

/\* --- write your Section 4 code here --- \*/

/\* =====  
 SECTION 5 | Scope and variable visibility

Translate each step into C code.  
 Place a breakpoint on Step 1.  
 Press F10 to step through every line including `{` and `}`.  
 Watch the Locals panel: the variable `val` will appear,  
 change its value, and disappear as you move through blocks.  
 Answer Q14-Q15 from what you observe.

Step 1. Declare an integer variable named `val` with value 10.  
           >>> LOCALS: note the value of `val` (POINT A)  
 Step 2. Open a code block {

```

Step 3.      Declare an integer variable named  val  with value 20.
            >>> LOCALS: note the value of val      (POINT B)
Step 4.      Open another code block {
Step 5.      Declare an integer variable named  val  with value 30.
            >>> LOCALS: note the value of val      (POINT C)
Step 6.      Close the inner block }
            >>> LOCALS: note the value of val      (POINT D)
Step 7.      Close the outer block }
            >>> LOCALS: note the value of val      (POINT E)
===== */

```

```

/* --- write your Section 5 code here --- */

```

```

/* =====

```

#### SECTION 6 | Logical and relational operators

Translate each step into one line of C code.  
Place a breakpoint on Step 1.  
Press F10 to step through all lines.  
Read ALL values from the Locals panel.  
Answer Q16-Q17 from what you observe.

```

Step 1.  Declare three integer variables on one line:
        lg_a = 5,  lg_b = 0,  lg_c = -3
Step 2.  Declare  int  res1  =  lg_a && lg_b
Step 3.  Declare  int  res2  =  lg_a || lg_b
Step 4.  Declare  int  res3  =  !lg_a
Step 5.  Declare  int  res4  =  !lg_b
Step 6.  Declare  int  res5  =  !!lg_a
Step 7.  Declare  int  res6  =  lg_c && lg_a
Step 8.  Declare  int  res7  =  ( lg_a > 3 )
Step 9.  Declare  int  res8  =  ( lg_a == lg_a )
===== */

```

```

/* --- write your Section 6 code here --- */

```

```

/* =====

```

#### SECTION 7 | Unsigned types, prefix and postfix

Translate each step into C code.  
Place a breakpoint on Step 1.  
Press F10 to step through every line.  
Read ALL variables in Locals at each step.  
Answer Q18-Q22 from what you observe.

*Part A - Unsigned wraparound:*

```
Step 1.  Declare unsigned int    u_i  = 0
Step 2.  Declare unsigned char   u_c  = 0
Step 3.  Declare unsigned short  u_s  = 0
Step 4.  Decrement all three on separate lines:
        u_i--;
        u_c--;
        u_s--;
        >>> LOCALS: read u_i, u_c, u_s after each decrement
Step 5.  Declare unsigned char   uc_max = 255
Step 6.  Increment: uc_max++;
Step 7.  Declare unsigned char   uc_over = 250
Step 8.  Add 10 :   uc_over = uc_over + 10;
```

*Part B - Prefix vs postfix:*

```
Step 9.  Declare int pp_i = 5
Step 10. Declare int pp_a = pp_i++
        >>> LOCALS after F10: read BOTH pp_a and pp_i
Step 11. Declare int pp_b = ++pp_i
        >>> LOCALS after F10: read BOTH pp_b and pp_i
===== */

/* --- write your Section 7 code here --- */

return 0;
}
```

## PART C — Canvas Quiz

22 questions · 1 pt each · no time limit · one attempt